

PREDICTIVE MAINTENANCE

OF INDUSTRIAL MACHINES

A comparative study of machine-learning algorithms on machine sensor data for early failure prediction.

Supervisor

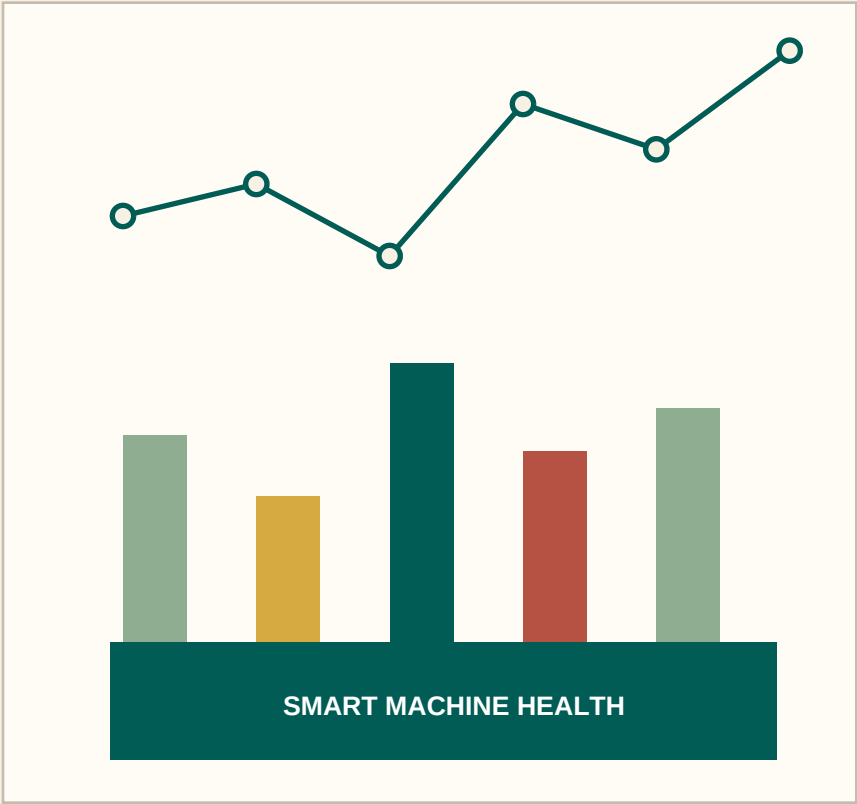
Department of Computer Science

Course

Machine Learning Project

Focus

Classification & Metrics



THE PROBLEM

Why maintenance needs prediction, not only repair

CORE IDEA

Industrial machines rarely fail without signs. Temperature, vibration, torque and tool wear often shift before breakdown.

- 1

Reactive maintenance
Repair happens after failure; downtime is already occurring.
- 2

Preventive maintenance
Scheduled service may replace parts too early or too late.
- 3

Predictive maintenance
ML estimates risk before failure and supports smarter planning.

RESEARCH QUESTION

Which machine-learning model can most reliably predict industrial machine failure using sensor features?

Sensors
6+

Target
Binary

Goal
Early warning

THE DATASET

Machine condition readings used for failure prediction

10,000

Rows / machine records

14

Attributes

6+

Key sensor features

Binary

Failure target

FEATURE GROUPS

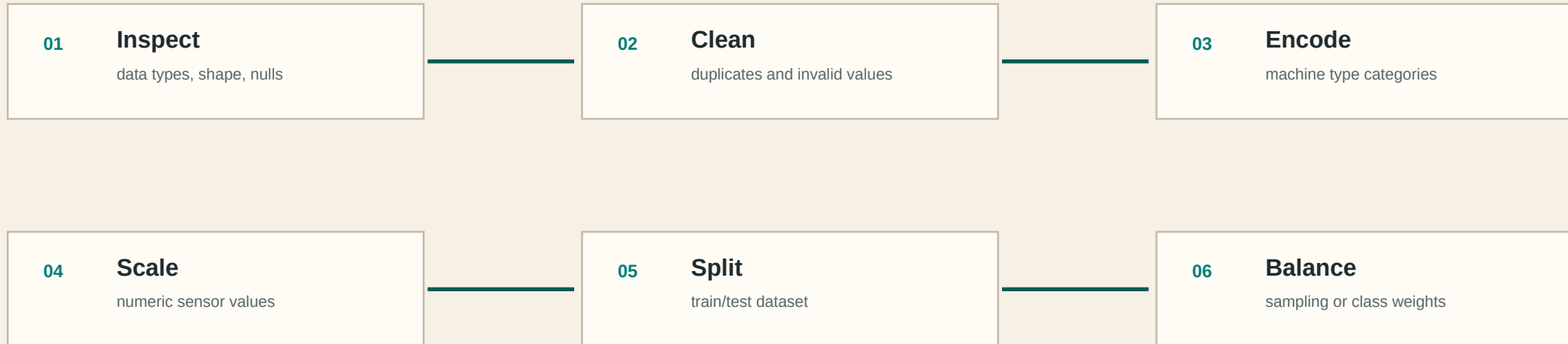
- Temperature
- Rotational speed
- Torque
- Tool wear
- Machine type

KEY CHALLENGE

Failure cases are usually rare, so accuracy alone can look high while the model still misses real failures.

DATASET PREPROCESSING

A clean pipeline before model training



WHY PREPROCESSING MATTERS

KNN and SVM are distance-sensitive, so scaling is required. Machine type must be encoded, and imbalanced failure classes need special attention.

PRECISION METRICS

Classification and regression evaluation

CLASSIFICATION METRICS

Confusion matrix

	Pred. Fail	Pred. OK
Actual Fail	TP	FN
Actual OK	FP	TN

Accuracy $(TP + TN) / \text{Total}$

Precision $TP / (TP + FP)$

Recall $TP / (TP + FN)$

F1-score $2PR / (P + R)$

REGRESSION METRICS

- MAE: average absolute error
- MSE / RMSE: squared error penalty
- R2: explained variance

In maintenance, recall matters because a missed failure is usually more serious than a false alarm.

LINEAR & LOGISTIC REGRESSION

The mathematical baseline

LINEAR REGRESSION

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Best for continuous predictions such as remaining useful life or expected wear.

CONTINUOUS OUTPUT

LOGISTIC REGRESSION

$$P(y=1) = 1 / (1 + e^{-z})$$

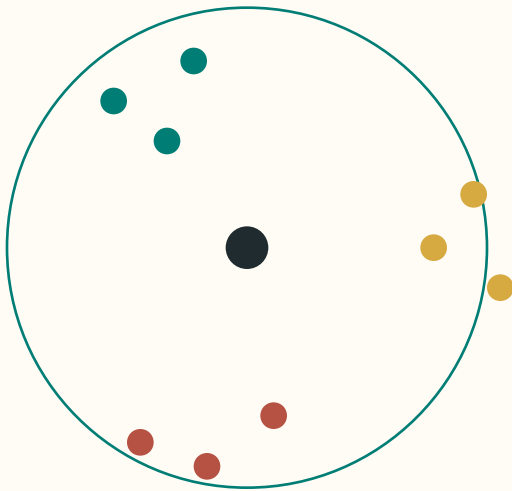
Best for binary failure prediction: fail vs. no fail.

PROBABILITY OUTPUT

MODEL 1 - K-NEAREST NEIGHBOR

Distance-based classification

K = 3 / 5 / 7



METHOD

- Store all training examples.
- Find the K nearest sensor patterns.
- Use majority voting for class label.
- Sensitive to scaling and K value.

$$d(x,y) = \text{sqrt}(\text{sum}((x_i - y_i)^2))$$

MODEL 2 - NAIVE BAYES

Probability-based classification

BAYES THEOREM

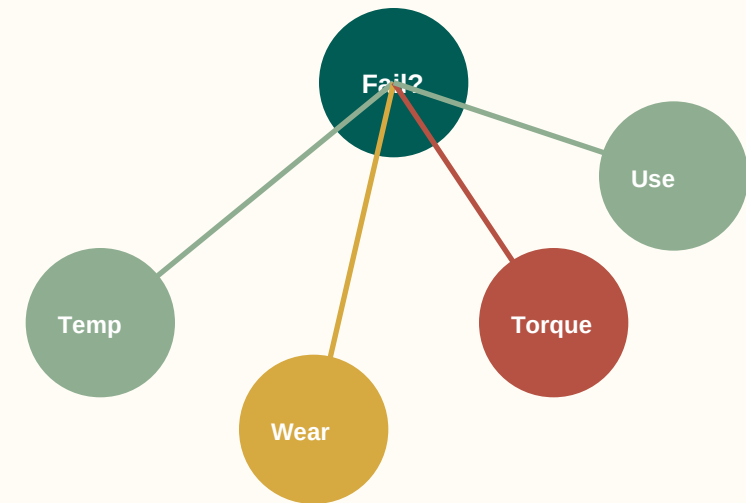
$$P(C|X) = P(X|C) * P(C) / P(X)$$

Where C is the class and X is the set of observed sensor features.

Assumption

Features are conditionally independent given the class.

PROBABILISTIC VIEW

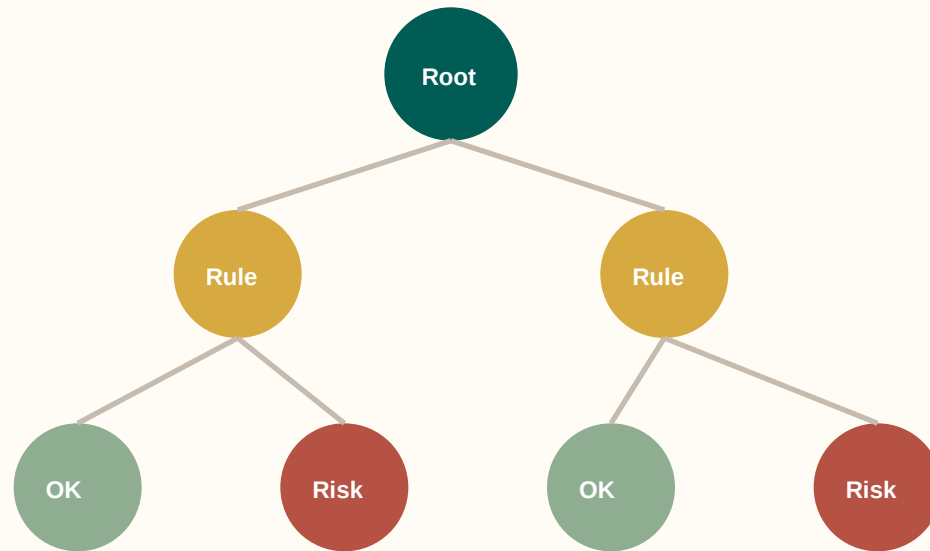


Use: Fast baseline for failure/no-failure classification.

DECISION TREE & SVM

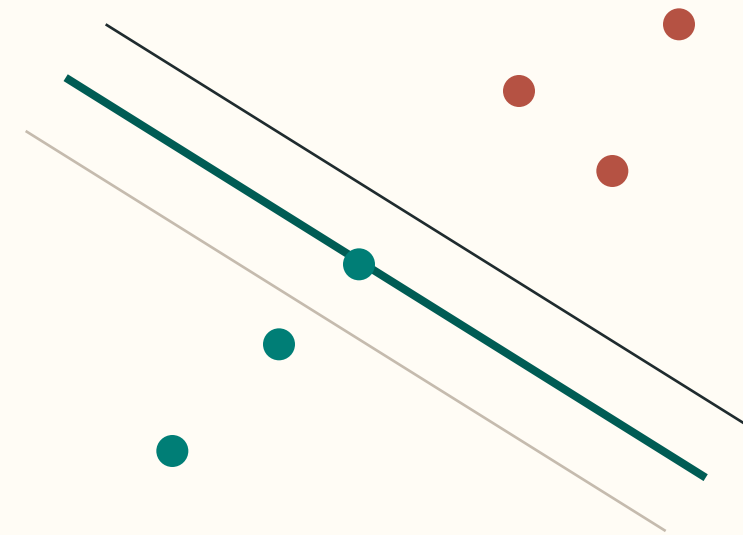
Interpretability vs. margin maximization

DECISION TREE



Splits data using feature conditions such as tool wear, torque or temperature.

SUPPORT VECTOR MACHINE



Finds a separating hyperplane and maximizes the margin between classes.

MODEL COMPARISON

Select model using balanced metrics, not accuracy alone

COMPARISON MATRIX

Model	Strength	Limitation	Best metric
KNN	Simple, local patterns	Needs scaling	F1 / Recall
Naive Bayes	Fast baseline	Independence assumption	Precision
Decision Tree	Explainable rules	Can overfit	Recall
SVM	Strong boundary	Tuning required	AUC / F1

Why recall matters

A false negative means a real failure is missed.

Accuracy warning

If failure cases are rare, accuracy can be misleading.

Best model choice

Use a balanced score: recall + precision + F1.

reliable detection + acceptable false alarms

CONCLUSION & FUTURE WORK

From model comparison to smart maintenance

CONCLUSION

- Predictive maintenance reduces downtime and improves planning.
- ML models can learn failure patterns from machine sensor readings.
- Preprocessing and evaluation metrics are essential for trustworthy results.
- Recall and F1-score should guide the final model decision.

FUTURE IMPROVEMENTS

MODELS

Random Forest, XGBoost, Neural Networks

EVALUATION

Cross-validation for robust comparison

FEATURES

Feature importance and sensor analysis

DEPLOYMENT

Real-time dashboard and IoT alerts

Final outcome: an ML-based maintenance decision-support system for smart industries.